

Probability

If we try an experiment N times
ex flip a coin

A is an event

let $A = \text{heads}$

$N(A) = \# \text{ of time getting heads}$

$$P(A) \approx \frac{N(A)}{N} \quad N \text{ very large}$$

$$P(A) = \lim_{N \rightarrow \infty} \frac{N(A)}{N}$$

Sample Space

Ω : sample space (set of all outcome)

ex Flip a coin $\Omega = \{H, T\}$

ex Throw a 6-sided die $\Omega = \{1, 2, 3, 4, 5, 6\}$

Outcome

$\omega \in \Omega$ is an outcome

ex roll a die, get 3
 outcome

event

subset of Ω

$\{1, 2, 3, 4, 5, 6\}$

ex Roll a die. Get a number ≤ 3

$$A = \{1, 2, 3\} \subset \Omega$$

Get an even number

$$E = \{2, 4, 6\}$$

Set theory

Union : "OR" $\cup$

$A \cup E$ Roll a die, get either ≤ 3 or even

$$A \cup E = \{1, 2, 3, 4, 6\}$$

Intersection : "AND" $\cap$ Roll a die, get both ≤ 3 and even

$$A \cap E = AE = \{1, 2, 3\} \cap \{2, 4, 6\} = \{2\}$$

Complement : "NOT" Everything in Ω not in E

$$E^c = \{2, 4, 6\}^c = \{1, 3, 5\} = \emptyset \quad \boxed{A \cup A^c = \Omega}$$

Not even

ex $(A \cup E)^c$ Not either ≤ 3 or even

$$= \{1, 2, 3, 4, 6\}^c$$

$$= \{5\}$$

$\Rightarrow$ neither ≤ 3 nor even

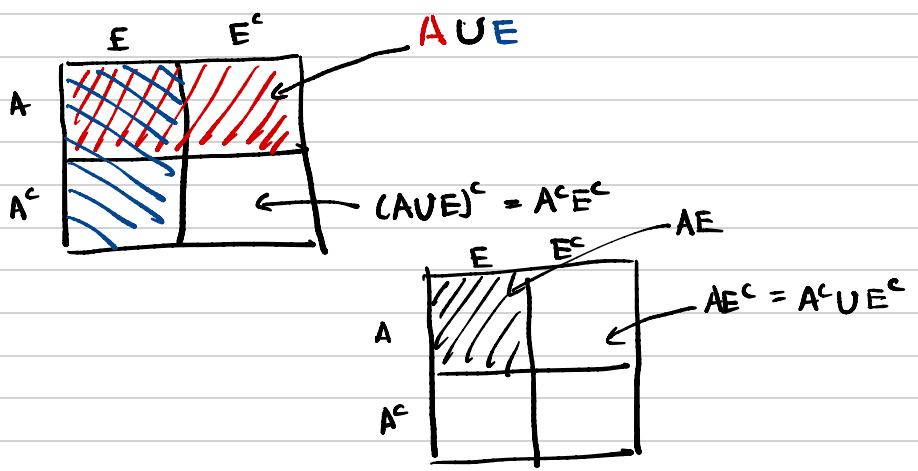
$\Rightarrow$ Not ≤ 3 AND NOT even

DeMorgan's Laws

$$(A \cup E)^c = A^c E^c$$

$$(AE)^c = A^c \cup E^c$$

Karnaugh Map K-map



	E	E ^c
A	2	1, 3
A ^c	4, 6	5

For now, let's assume all outcomes are equally likely

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$P(A) = \frac{|A|}{|\Omega|}$$

Cardinality

Size of set

ex $A = \{1, 2, 3\}$

$$|A| = 3$$

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$|\Omega| = 6$$

$$P(A) = \frac{|A|}{|\Omega|} = \frac{3}{6} = \frac{1}{2}$$

$$P(E) = \frac{1}{2}$$

$$P(\{3\}) = \frac{1}{6}$$

$$P(A \cup E) = \frac{5}{6}$$

$$P(AE) = \frac{1}{6}$$

ex Ten balls, # 1-10 in an opaque bag

Draw 3 balls, without replacement

$$\Omega = \{(\omega_1, \omega_2, \omega_3) : \omega_1, \omega_2, \omega_3 \in \{1, 2, 3, \dots, 10\} \\ \omega_1 \neq \omega_2 \neq \omega_3\}$$

$$|\Omega| = 10 \times 9 \times 8 = 720$$

$$(1, 2, 3) \quad (3, 2, 1)$$

Powerball

1-69 white ball

Draw 5 white, w/o replace

1-26 red ball

Draw 1 red

\$1.7 billion

\$1,700,000,000

\$2 ticket

Probability Space

Triplet $(\Omega, \mathcal{F}, P)$

Ω : sample space

$\omega \in \Omega$: outcome

A, E, etc.

$\mathcal{F}$: event space

set of all events

Event = subset of Ω

$$|\mathcal{F}| = 2^{|\Omega|}$$

$\Rightarrow$ set of all subsets of Ω

ex Thrown a die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$|\Omega| = 6$$

$$\mathcal{F} = \{ \{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}$$

$$\{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{1, 6\}$$

$$\{2, 3\}, \{2, 4\}$$

$$\{6\}$$

$$5$$

$$4$$

$$3$$

$$2$$

$$1$$

$$15$$

$$\{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}, \{1, 2, 6\}$$

$$\{1, 3, 4\}, \{1, 3, 5\} \dots$$

$$\{1, 2, 3, 4, 5, 6\} = \Omega$$

$$\Omega^c = \emptyset$$

$$1$$

$$1$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$\binom{6}{1} = \frac{6!}{1!5!} = 6$$

$$\binom{6}{2} = \frac{6!}{2!4!} = \frac{30}{2} = 15$$

$$\binom{6}{3} = \frac{6!}{3!3!} = \frac{120}{6} = 20$$

$$\binom{6}{4} = \frac{6!}{4!2!} = 15$$

$$= \underset{21}{6} + \underset{41}{15} + \underset{56}{20} + \underset{62}{15} + 6 + 1 + 1 = 64$$

P: Probability measure on $\mathcal{F}$

Event
space

$\mathcal{F}$ must satisfy $\exists$ Axiom
===== just accept it!
no proof.

E1: $\Omega \in \mathcal{F}$

E2: If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$
closed under complements

E3: $A_1, A_2 \in \mathcal{F}$, then $(A_1 \cup A_2) \in \mathcal{F}$
closed under unions
 $A_1, A_2, A_3, \dots \in \mathcal{F}$, then $\bigcup_i A_i \in \mathcal{F}$

Derived properties

$\emptyset$ = null set

e4: $\emptyset \in \mathcal{F}$

Proof: $\Omega \in \mathcal{F}$ (E1)
 $\Omega^c \in \mathcal{F}$ (E2)
 $\emptyset \in \mathcal{F}$ $\square$

e5: $A_1, A_2 \in \mathcal{F}$, then $A_1 A_2 \in \mathcal{F}$

Proof: $A_1, A_2 \in \mathcal{F} \rightarrow A_1^c, A_2^c \in \mathcal{F}$ (E2)
 $\rightarrow (A_1^c \cup A_2^c) \in \mathcal{F}$ (E3)
 $\rightarrow (A_1^c \cup A_2^c)^c \in \mathcal{F}$ (E2)
 $\rightarrow ((A_1 A_2)^c)^c = A_1 A_2 \in \mathcal{F}$ $\square$

e6: $A_1, A_2, A_3, \dots \in \mathcal{F} \rightarrow \bigcap_i A_i \in \mathcal{F}$

Use induction

$A_1, A_2 \in \mathcal{F} \rightarrow A_1 \cap A_2 \in \mathcal{F}$

$\Rightarrow (A_1 \cap A_2) \cap A_3 \in \mathcal{F}$

$\Rightarrow \dots$ etc

Probability
Measure
P

P is a measure on $\mathcal{F}$

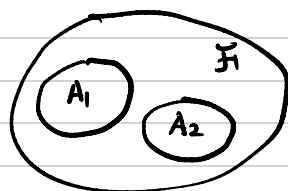
Must satisfy 3 Axioms

P1. $P(A) \geq 0 \quad \forall A \in \mathcal{F}$

P2. $A_1, A_2 \in \mathcal{F}$ and $\underline{A_1 \cap A_2 = \emptyset}$

A_1, A_2 mutually exclusive

$\Rightarrow P(A_1 \cup A_2) = P(A_1) + P(A_2)$



P3. $P(\Omega) = 1$

derived properties

p4. $P(A^c) = 1 - P(A)$

Proof. $A, A^c \in \mathcal{F}$ and $A \cap A^c = \emptyset$
 $\in \mathcal{F}$ def. of complement

P2 $P(A) + P(A^c) = P(\underline{A \cup A^c})$
 Ω def. of complement

$P(A) + P(A^c) = \underline{P(\Omega) = 1}$
P3

$P(A^c) = 1 - P(A)$ $\square$

P5. $P(A) \leq 1$

Proof: $P(A) = 1 - \underline{P(A^c)}$ p4
 $P(A^c) \geq 0$ P1

$P(A) \leq 1$ $\square$

p6. $P(\emptyset) = 0$

Proof: (P3) $P(\Omega) = 1$

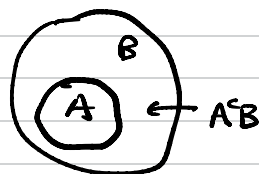
$\Omega^c = \emptyset$

$P(\emptyset) = P(\Omega^c) = 1 - P(\Omega) = 1 - 1 = 0$

□

p7. $A \subset B \Rightarrow P(A) \leq P(B)$

↑
is a subset of



$A = AB$

Proof: $B = A \cup (A^c B)$

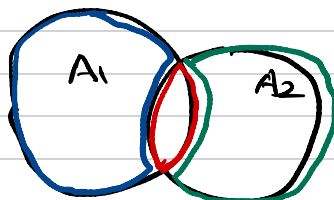
$AA^c B = \emptyset$ (P2)

$P(B) = P(A) + \underbrace{P(A^c B)}_{\geq 0}$ (P1)

$\Rightarrow P(B) \geq P(A)$ □

p8. $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 A_2)$

Proof



$A_1 \cup A_2 = A_1 A_2^c \cup A_1 A_2 \cup A_1^c A_2$

$P(A_1 \cup A_2) = P(A_1 A_2^c) + P(A_1 A_2) + P(A_1^c A_2)$

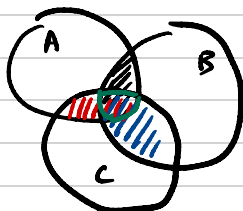
$= \underbrace{P(A_1 A_2^c) + P(A_1 A_2)}_{= P(A_1)} + \underbrace{P(A_1^c A_2) + P(A_1 A_2)}_{= P(A_2)} - \underbrace{P(A_1 A_2)}_{= 0}$

$= P(A_1) + P(A_2) - P(A_1 A_2)$ telescoping

□

p9. $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(BC) - P(AC) + P(ABC)$

Proof



$$\begin{aligned} P(A \cup B \cup C) &= P[(A \cup B) \cup C] \\ &= \underline{P(A \cup B)} + P(C) - P[(A \cup B)C] \\ &= P(A) + P(B) - P(AB) + P(C) - P(AC \cup BC) \end{aligned}$$

$$= P(A) + P(B) - P(AB) + \underline{P(C)} - [P(AC) + P(BC) - P(ABC)]$$

$$= P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$$

QED

ex

$$P(A) = 0.6$$

$$P(B) = P(C) = 0.3$$

$$P(B^c C) = P(AB^c C^c) = 0.2$$

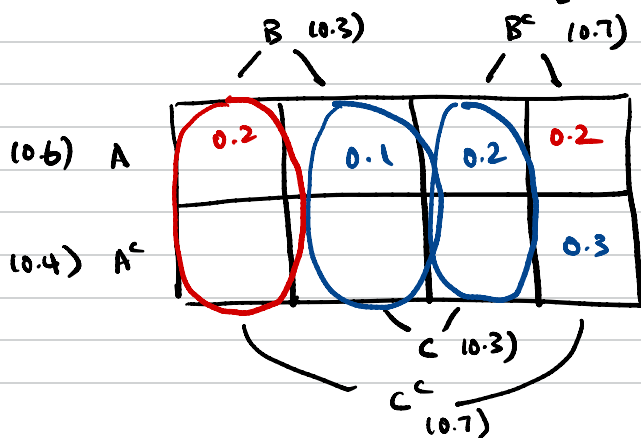
$$\text{Find } P(BC) = 0.1$$

$$P(BC^c) = 0.2 ?$$

$$P[(A \cup B \cup C)^c]$$

$$= P[A^c \cap B^c \cap C^c]$$

$$= 0.3$$



4

$$P(A) = P(B) = P(C) = 0.3$$

$$P(ABC) = 3P(ABC)$$

$$P(A \cup B \cup C) = P(B \cup C) = 0.5$$

$$P(A^c B^c C^c) = 0.48$$

Find P of all events in $\mathcal{F}$

		B		B ^c	
		0.3		0.7	
0.3	A	2x 0.18	AB x=0.09	0.01	0.02
0.7	A ^c	0.02	0.01	0.19	0.48
		C		C ^c	
		0.3		0.7	

$$2x + 0.02 + 0.02 + 0.48 = 0.7$$

$$2x + 0.52 = 0.7$$

$$2x = 0.18$$

$$x = 0.09$$

$$0.3 - (0.18 + 0.09 + 0.02) = 0.01$$

$$0.7 - (0.02 + 0.01 + 0.48) = 0.19$$

$$0.7 - (0.19 + 0.02 + 0.48) = 0.01$$

Powertball

1-69 white balls

Draw 5 white, w/o replace

1-26 red balls

Draw 1 red

\$1.7 billion

\$1,700,000,000

\$2 ticket

What's the probability of winning?

$$\frac{69 \times 68 \times 67 \times 66 \times 65}{5 \times 4 \times 3 \times 2 \times 1} \times 26 = \frac{1348621560}{120} \times 26$$
$$= 11238513 \times 26$$
$$= 292,201,338$$

(1, 2, 3, 4, 5)

(5, 4, 3, 2, 1)

(5, 1, 2, 3, 4)

⋮

etc

292.201338 million

$$P(\text{winning}) = \frac{1}{292,201,338}$$

$$\$2 \rightarrow \$1,700,000,000$$

$$\frac{2}{1,700,000,000} < P(\text{winning}) \Rightarrow \text{good bet.}$$

$$\frac{1,700,000,000}{292,201,338} \approx 2 \quad \checkmark$$
$$= 5.81771$$

Counting

n objects
 m choices for each object

$\Rightarrow m \times n$ total choices

ex license plate 3 letters + 4 digits.
How many license plate #?

$L L L \# \# \# \#$
 $26 \ 26 \ 26 \ 10 \ 10 \ 10 \ 10$

$$26 \times 26 \times 26 \times 10 \times 10 \times 10 \times 10 = 175,760,000$$

ex What is $P(A)$

A = event where no repeat letters or #'s

ABC1234 ✓

AAC1234 ✗

ABC1134 ✗

$|A|$

$$175,760,000$$

$$= \frac{78,624,000}{175,760,000}$$

$$\approx 0.4473$$

$$|A| = 26 \times 25 \times 24 \times 10 \times 9 \times 8 \times 7 \\ = 78,624,000$$

NOTE: ordering of letters and numbers matter.

ex How many license plates contains A, B, C

1, 2, 3, 4

ABC1234

ABC1243

....

of way to order A, B, C
×

of way to order 1, 2, 3, 4

$$3 \times 2 \times 1 = 3! = 6 \quad \text{ways to order ABC}$$

$$4 \times 3 \times 2 \times 1 = 4! = 24$$

ways to order 1234

$$6 \times 24 = 144 \text{ ways}$$

Permutation

Ways of ordering n distinct objects
 $n!$ ways.

AAC 1114